Application of data fragmentation and replication methods in the cloud: a review

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***Abstract*—Data fragmentation and replication are an essential issue in cloud databases, since information demands are huge and improving performance through distributed database design techniques is a necessity. This paper aims to provide a review of the literature about data fragmentation and replication methods applied in a cloud environment in the last eight years, to determine if the methods take into account both techniques (fragmentation and replication) together, consider a cloud database, are easy to implement, are focused on improving the performance of the database, are complete (i.e., the paper includes all the necessary information to implement the method), and if they provide a cost model. To meet this objective we analyzed and classified 83 papers concerning data fragmentation and replication techniques. Finally, we selected the best method of our comparative analysis and presented an architecture of a web application for data fragmentation and replication in the cloud based on such method.**

Keywords—Data fragmentation; replication; cloud environment.

I. Introduction

One of the main issues faced in cloud computing is data retrieval [1]. In this context, data is stored on multiple dynamic virtual servers across the cloud. Before being distributed to multiple servers, the data is fragmented. Existing data fragmentation techniques are aimed at improving the data manipulation process, decreasing processing time, facilitating data manipulation, optimizing storage, increasing flexibility, distributing processing costs and facilitating data distribution and transportation [2]. In addition, data replication allows reducing user waiting time, speeding up data access and increasing availability by providing the user with different replicas of the same service, all of them with a coherent state [3]. Currently, several data fragmentation and replication techniques for the cloud have been proposed. Nevertheless, to the best of our knowledge, there is not a review paper of these works. The objective of this paper is to perform a comparative analysis between the different data fragmentation and replication techniques applied in the cloud to select a technique for its future implementation, which must comply with: a completeness in its description, that attends both the fragmentation and the replication, that contains a cost model that focuses on the cloud, that its implementation is easy and that it improves the performance of the database. We also present the architecture of a web application for data fragmentation and replication in the cloud based on the selected method. This review gives relevant information to practitioners and researchers about the main characteristics of the fragmentation and replication techniques applied in the cloud. The rest of this work is organized as follows: Section II discusses the analysis methodology and the classification of the articles, Section III presents the architecture of the web application for data fragmentation and replication in the cloud, finally, Section IV gives the conclusions and future work.

II. Analysis

In the following subsections an in-depth analysis of the state of the art is carried out. The first subsection includes the methodology used for the selection and analysis of each article. The second subsection addresses a classification of articles considering the publisher and the year. The third subsection shows an analysis of the articles focused on fragmentation and replication. Finally, in the fourth subsection, the main database management systems that are mentioned in some articles are shown.

A. Research methodology

In order to carry out this work, all the related articles published by the main scientific editorials were searched using a methodology for their subsequent classification. Figure 1 shows the methodology used for the selection and analysis of each article.

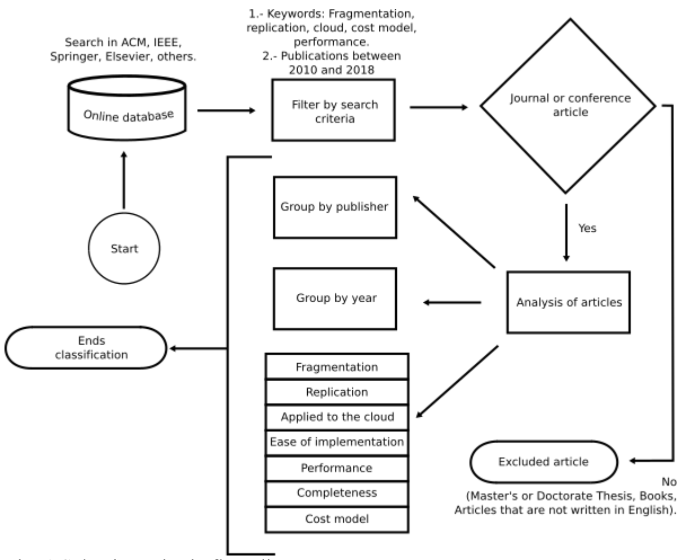

Fig. 1 Selection criteria flow diagram

The methodology is composed of three stages. The first stage was searching in the major databases of electronic journals for a comprehensive bibliography of relevant research of data fragmentation and replication in the cloud. The main digital libraries considered were: 1) ACM digital library, 2) IEEE Xplore Digital Library, 3) Science Direct (Elsevier), and 4) Springer Link. Papers published by academic journals, workshops, and international conference proceedings are thought to be reliable and worthy. As a second stage, we employed a keyword-based search to select the most relevant articles. The main keywords were: 1) Fragmentation or partitioning, 2) Replication, 3) Cloud, 4) Cost model, and 5) Performance. The articles that fulfilled these characteristics were selected and the works not suitable for the study were discarded. The following statements describe the criteria considered for the omission of a research paper:

- Unpublished working papers, non-peer-reviewed papers, non-English papers, textbooks, Master and Doctoral dissertations.
- Because research of data fragmentation and replication in the cloud is relatively current, we have only searched recent articles published between 2010 and 2018.

The selection process resulted in 83 papers, which were analyzed by seven characteristics to register if they fulfilled them and in which way they did it, for example, we classified the works by the problem addressed, i.e., fragmentation, replication or both techniques together. The characteristics are: 1) Fragmentation, 2) Replication, 3) Applied to the cloud, 4) Ease of implementation, 5) Performance, 6) Completeness, and 6) Cost model. In this step the articles were also classified by year and by editorial. Tables 1 to 5 describe the record of each article and the analysis of each characteristic.

Article	1	2	3	4	5	6	7
Abdel et al. [4]	x	x	x	x	x	x	x
Thomson et al. [5]		x			x	x	x
López et al. [6]		x	x		x	x	x
Marcus et al. [7]	x	x	x	x	x		x
Gudadhe et al. [8]		x	x	x	x		
Dhamane et al. [9]		x	x	x	x	x	
Shwe et al. [10]		x	x	x	x		
Bollwein et al. [11]	x		x		x	x	
Xiao et al. [12]		x	x	x	x	x	
Abdul et al. [13]		x	x	x	x	x	x
1) Fragmentation, 2) Replication, 3) Applied to the cloud, 4) Ease of implementation, 5) Performance, 6) Completeness, 7) Cost model							

Table 1: Comparison of related ACM works

Article	1	2	3	4	5	6	7
Nuaimi et al. [14]	x	x	x	x	x		
Stiemer et al. [15]		x	x			x	x
Tziritas et al. [16]		x	x	x	x	x	x
Salunkhe et al. [17]	x	x	x	x			
Ali et al. [18]		x	x		x	x	x
Nuaimi et al. [19]		x	x	x	x	x	
Tseng et al. [20]		x	x	x	x	x	
Zhu et al. [21]		x	x	x		x	
Moral et al. [1]		x	x		x		
Mseddi et al. [22]		x	x	x	x	x	
Thara et al. [23]		x	x	x	x		
Hou et al. [24]		x	x	x	x	x	
Li et al. [25]		x	x		x		
Tripathi et al. [26]		x	x	x	x	x	
Tian et al. [27]		x	x	x		x	
Luo et al. [28]		x	x		x	x	
Xie et al. [29]		x	x	x	x	x	x
Hsu et al. [30]		x	x	x	x	x	

Table 2: Comparison of related IEEE works

Article	1	2	3	4	5	6	7
Alami et al. [31]		x	x	x	x		
Hamrouni et al. [32]		x			x	x	
Kaur et al. [33]		x	x		x	x	x
Mansouri et al. [34]		x	x	x	x	x	x
Amer et al. [35]	x	x		x	x	x	x
Brigit [36]	x	x		x	x	x	x
Montoya et al. [37]		x	x		x		
Mansouri et al. [38]		x	x	x	x	x	x
Rosas et al. [39]	x		x	x	x	x	
Fan et al. [40]	x		x	x	x		
Gkatzikis et al. [41]		x	x	x	x	x	x

Article	1	2	3	4	5	6	7
Long et al. [42]		x	x		x		x
Sengupta et al. [43]	x	x	x	x		x	x
Zeng et al. [44]		x	x	x	x	x	x
Shabeera et al. [45]			x		x	x	x
Habib et al. [46]				x	x		
Sánchez et al. [47]	x		x	x		x	
Chen et al. [48]		x	x	x	x	x	
Xia et al. [49]			x	x	x	x	x
Romero et al. [50]	x		x	x	x	x	x
1) Fragmentation, 2) Replication, 3) Applied to the cloud, 4) Ease of implementation, 5) Performance, 6) Completeness, 7) Cost model							

Table 3: Comparison of related Elsevier works

Article	1	2	3	4	5	6	7
Benkrid et al. [51]	x	x	x		x	x	x
Mansouri et al. [52]		x	x	x	x	x	x
Guabtni et al. [53]			x	x	x	x	
Darabant et al. [54]		x	x	x	x	x	x
Mansouri [55]		x	x	x	x	x	x
Edwin et al. [56]		x	x	x	x	x	x
Abdel et al. [57]	x	x	x		x		
Wiese [58]	x	x	x	x	x		
Grolinger et al. [59]	x	x	x		x		
Mohammad et al. [60]	x	x	x		x		
Abdel et al. [61]	x	x	x	x	x	x	x
Abdel et al. [62]	x	x	x	x	x		
Kish et al. [63]	x	x	x	x	x		x
Zou et al. [64]	x	x	x	x	x		
Insaf et al. [65]		x	x		x		
Gao et al. [66]	x	x			x	x	
Rehman et al. [67]		x	x	x			
Bellatreche et al. [68]	x		x	x		x	
Ahirrao et al. [69]	x		x	x	x	x	
Hassen et al. [70]	x	x		x	x	x	
Lehner et al. [71]	x	x	x	x	x		

Table 4: Comparison of related Springer works

Article	1	2	3	4	5	6	7
Phaphoom et al. [72]		x	x	x	x		
Wiese et al. [73]	x	x	x	x	x	x	x
Noraziah et al. [74]	x	x	x	x			
Fauzi et al. [75]	x	x	x	x			
Mrabti et al. [76]	x	x	x	x	x	x	
Khandelwal et al. [77]	x	x	x	x			
Singh et al. [78]		x	x	x			
S et al. [79]	x		x	x		x	x
Scheidgen et al. [80]	x			x	x		x
Hauglid et al. [81]	x	x		x	x	x	x
Das et al. [82]	x	x	x	x	x	x	
Wiese [83]	x	x		x			
Bermbach et al. [84]		x	x	x	x		
Barsoum et al. [85]	x		x	x	x	x	

Table 5: Comparison of related others works

It is observed in the tables 1 to 5 the comparison of all the works found in the main scientific digital libraries, which addressed most of the topics of interest. The selected articles were evaluated in each item using a methodology to determine if each item meets the characteristics. From all the analyzed works it is observed that few articles fulfill all the desired qualities. The articles [4], [61] and [73] comply with all the characteristics.

B. Classification considering editorial and year

Figure 2 shows a bar graph that relates the number of articles with the publisher; most articles are concentrated in the Springer publishing house. Springer articles addressed mostly data replication ([51], [52], [54]-[67], [70], [71]), 18 of 21 articles. 12 papers considered both fragmentation and replication together ([51], [57]-[64], [66], [70] and [71]). The editorials were evaluated in relation to the proximity of the articles with the desired characteristics, Springer ranked second. ACM is the editorial with the highest rating and IEEE the lowest. Figure 3 shows the results of the proximity evaluation with the topics of interest of each editorial.

The distribution of the analyzed papers according to their year of publication is depicted in Figure 4. It is observed that in the year 2017 there was an increase in the number of articles. It is noted that of the 18 articles only two belong to Springer ([54] and [56]), the publisher with the largest number of articles in the previous graph, seven belong to Elsevier ([35], [37], [38], [41], [45], [47] and [49]), four to ACM ([6], [8], [10] and [11]), four to IEEE ([16], [25], [29] and [30]) and one to the category "others" ([73]). In 2013, the number of articles of the previous three years was doubled, since the number of works that consider fragmentation and replication together increased four times more ([81] in the previous three years; [59], [71], [74], and [77] in 2013). In 2017, 12 articles out of 18 addressed data replication ([6], [8], [10], [16], [25], [29], [30], [38], [41], [54], [56], and [73]).

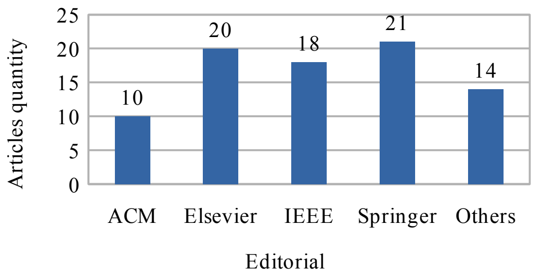

Fig. 2 Articles quantity per editorial

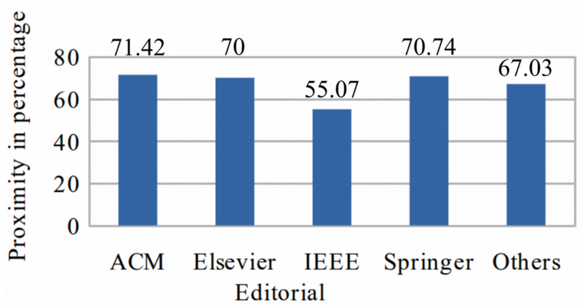

Fig. 3 Proximity to the topics of interest of each editorial

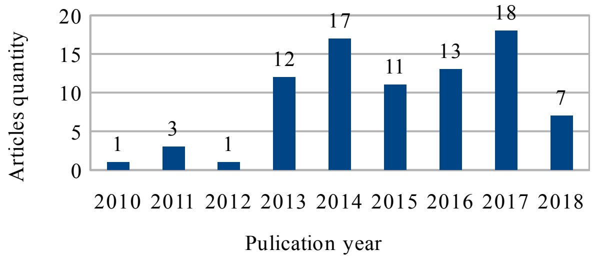

Fig. 4 Articles quantity per year

C. Fragmentation and replication

In Figure 5 there is a comparison between the amount of papers that consider data fragmentation ([4], [7], [11], [14], [17], [35], [36], [39], [40], [43], [47], [50], [51], [57]-[64], [66], [68]-[71], [73]-[77], [79]-[83], and [85]) and the number of works that take into account data replication ([4]-[10], [12]-[38], [41]-[44], [48], [51], [52], [54]-[67], [70]-[78], and [81]-[84]). 27 articles are distributed between the two topics that contain both themes together. In Figure 5 it is observed that replication is a more common topic among the 83 articles.

Figure 6 shows a bar graph that relates the number of articles by type of fragmentation. The number of articles that include fragmentation are 37, as shown in Figure 5, however there are articles that include more than one type of fragmentation ([59], [68], [70], [71], [73]-[75], and [77]), which is why in the graph of Figure 6 the total number of works is not equal to the total of fragmentation articles. Most articles that include more than one type of fragmentation do not fall into the category "Hybrid", because each type uses it separately or only contemplate those types in their work to solve other problems ([59], [68], [71], [73]-[75], and [77],). In the category "Other" are all articles that speak of some type of fragmentation that is not any of the other categories ([7], [14], [17], [36], [39], [43], [47], [63], [64], [66], [69], [76], [79], [80], and [85]).

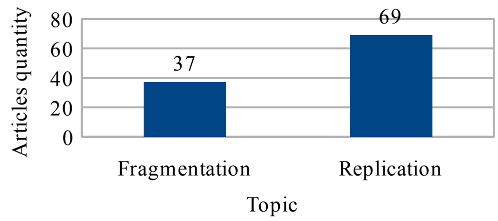

Fig. 5 Articles quantity per topic

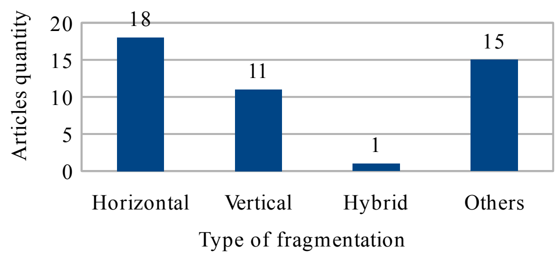

Fig. 6 Articles quantity per type of fragmentation

D. Database management systems

In Figure 7 the number of works per database management system (DBMS) is described. Most of the articles do not mention or do not use DBMSs to carry out their research. Figure 7 shows that the four most used DBMSs are PostgreSQL, Hbase, MySQL and Microsoft SQL Server. Some articles include more than one system, as [9], because it uses MySQL and PostgreSQL. Table 6 shows each article included in Figure 6 and to which system it is related.

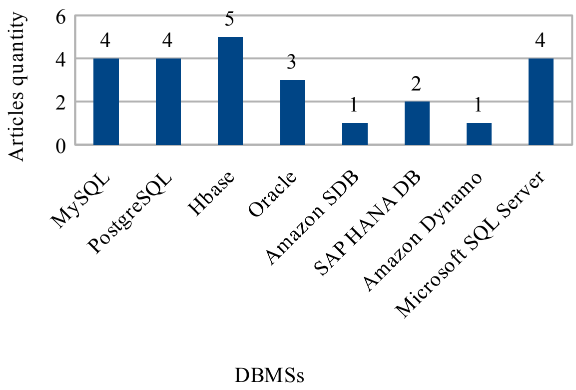

Fig. 7 Articles quantity per database management system

DBMSs	Articles
Mysql	[9], [53], [56], [64]
PostgreSQL	[7], [9], [11], [58]
Hbase	[12], [46], [50], [69], [79]
Oracle	[24], [68], [70]
Amazon Simple DB	[69]
SAP HANA DB	[73], [82]
Amazon Dynamo	[83]
Microsoft SQL Server	[57], [61], [62], [72]

Table 6: Articles per database management system

III. ARCHITECTURE

The three articles ([4], [61] and [73]) that involve fragmentation and replication, are applied to the cloud, are easy to implement, are focused on performance, have completeness and include a cost model, were evaluated and two of them were discarded. In [4], it is necessary to evaluate the relationships with respect to use cases that the same user provides, for this, more information is needed than what the user would like to capture in the proposed system. In a simulated environment, the work in [4] would be developed properly, but in a real environment it would not fulfill what was expected. In [73] to carry out the fragmentation authors use flexible queries, a topic that is not included in the delimitation of this research.

The chosen article, [61], presents a desired methodology and an easy implementation of the technique to include the costs in the allocation and replication. Figure 8 depicts the workflow proposed by [61].

Fragmentation and replication in [61] is based on the cost of creating, reading, updating and deleting predicates included in a matrix called MCRUD (Creation, Reading, Updating and Deletion Matrix), this matrix records each operation of a predicate in each site, in this way, obtaining the cost of each operation, determines how the relationship will be fragmented, where the fragments will be assigned and in which site they will be replicated.

Figure 8 shows the workflow that consists of six main steps. The first step is the analysis of the information of the queries extracted from the log previously provided by the user. In this first step, the MCRUD is constructed, which is obtained by placing the actions of creation, reading, updating and deletion for each site with respect to the predicate of the queries. The second step is creating the ALP (Attribute Locality Precedence) matrix obtained through the cost of each site. The cost of each site is determined by the cost functions described in [61]. Once the ALP matrix is built, in the third step, the horizontal fragmentation scheme is created according to the attribute with the highest operating cost.

The fourth step is obtaining a table with the final results of allocation and replication, where it is shown for each fragment, the site where it will be assigned and the site where it will be replicated. This information is determined with the MCRUD matrix, each fragment is assigned to the site where it has the highest cost of operations and is replicated in the site where more reading operations are performed. The fifth step is to apply fragmentation, allocation and replication according to the schemes described. The sixth step shows the result of this workflow, which is each site with the fragments and replicas.

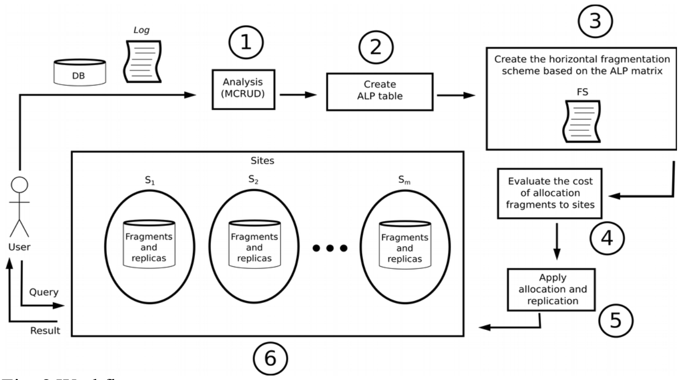

Fig. 8 Workflow

Figure 9 represents the architecture that was chosen. The Model View Controller (MVC) architecture was selected: In the view are the pages that interact with the user to request information about the database and how to access it. The managed JSF (Java Server Faces) beans that manage the information flow are located in the controller. JSF [86] was selected due to the ease of development for web applications with this framework and because it is free to use. In the model, the negotiation rules are defined, which in this particular scenario will be the application of the selected fragmentation and replication technique, besides contemplating the calculation of the cost model of the transformation carried out for a given database.

To carry out the validation of the selected technique, a web application will be built to perform fragmentation and replication automatically. The application will be implemented in AWS (Amazon Web Services) [87] because it provides reliable, scalable and inexpensive cloud computing services, and because it has the best accessibility to the chosen language. NetBeans [88] was chosen as IDE (Integrated

Development Environment) because of its intuitive interface and its great facility to develop applications with the selected framework. UWE (UML-Based Web Engineering) [89] was selected as a methodology for software development, since it provides specific steps aimed at web applications and its great relationship with UML (Unified Modeling Language) diagrams that are considered standards in the OMG (Object Management Group). MySQL [90] was chosen as DBMS, due to the speed in its queries and because it is one of the DBMSs most used in research papers according to Figure 7.

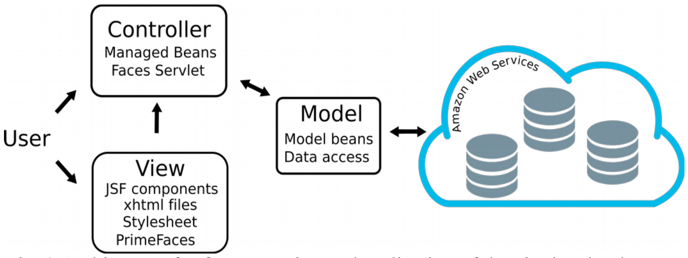

Fig. 9 Architecture for fragmentation and replication of data in the cloud

IV. CONCLUSIONS AND FUTURE WORK

The fragmentation and replication of data in the cloud is a topic widely used in the industry, and provided by various platforms. In this paper, we reviewed research papers of data fragmentation and replication methods applied in the cloud from 2010 to 2018. We conclude that only three works ([4], [61] and [73]) fulfilled the seven criteria of our comparative analysis. In addition, we classified the papers by editorial, year of publication, type of problem addressed, type of fragmentation, and database management system used to implement the techniques.

In the future, the web application will be implemented with the proposed architecture and will be validated with two case studies, one of them will be applied in CloudSim [91] and the other in AWS. When the application will be in development, the CloudSim platform will be used and the number of sites will not be limited, however when the application will be finished it will be hosted in AWS and there will only be three available sites due to the price of the same.

V. ACKNOWLEDGMENT

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VI. REFERENCES

[1] W. Delishiya Moral and B. M. Kumar, "Improve the data retrieval time and security through fragmentation and replication in the cloud," 2016 International Conference on Advanced Communication Control and Computing Technologies (ICACCCT), Ramanathapuram, 2016, pp. 539-545.

[2] A. Hudic, S. Islam, P. Kieseberg and E. R. Weippl, “Data confidentiality using fragmentation in cloud computing”, Int. J. Communication Networks and Distributed Systems, vol. 1, No. ¾, 2012, pp. 1-10.

[3] D. W. Sun, S. Gao, L. Z. Jin and X. W. Wang, “Modeling a dynamic data replication strategy to increase system availability in cloud computing environments”, Journal of Computer Science and Technology, vol. 27, No. 2, 2012, pp. 256-272.

[4] A. E. Abdel, N. L. Nadr and M. F. Tolba, “An Enhanced CRUD For Vertical Fragmentation Allocation And Replication Over The Cloud Environment”, INFOS '16: Proceedings of the 10th International Conference on Informatics and Systems, Giza, EG, 2016, pp. 146-152.

[5] A. Thomson, T. Diamond, S. Weng, K. Ren P. Shao and D. J. Abadi, “Fast Distributed Transactions and Strongly Consistent Replication for OLTP Database Systems”, ACM Transactions on Database Systems, Vol. 39, No. 2, 2014, pp. 11-50.

[6] C. López, R. Heinsen and E. Huh, “Improving Availability Applying Intelligent Replication in Federated Cloud Storage Based on Log Analysis”, International Conference on Machine Learning and Soft Computing 2017, Ho Chi Minh City, VN, 2017, pp. 148-153

[7] R. Marcus, O. Papaemmanouil, S. Semenova and S. Garber, “NashDB: An End-to-End Economic Method for Elastic Database Fragmentation, Replication, and Provisioning”, Proceedings of the 2018 International Conference on Management of Data, Houston, TX, USA, 2018, pp. 1253-1267.

[8] M. B. Gudadhe and A. J. Agrawal, “Performance Analysis Survey of Data Replication Strategies in Cloud Environment”, Proceedings of the 2017 International Conference on Big Data Research, Osaka, JP, 2017, pp. 38-43.

[9] R. Dhamane, M. Patiño, V. Vianello and R. Jiménez, “Performance Evaluation of Database Replication Systems”, Proceedings of the 18th International Database Engineering & Applications Symposium, Porto, PT, 2014, pp. 288-293.

[10] T. Shwe and M. Aritsugi, “Proactive Re-replication Strategy in HDFS based Cloud Data Center”, Proceedings of the10th International Conference on Utility and Cloud Computing, Austin, TX, USA, 2017, pp. 121-130.

[11] F. Bollwein and L. Wiese, “Separation of Duties for Multiple Relations in Cloud Databases as an Optimization Problem”, Proceedings of the 21st International Database Engineering & Applications Symposium, Bristol, UK, 2017, pp. 98-107.

[12] L. Xiao, K. Ren, Q. Zheng and G. A. Gibson, “ShardFS vs. IndexFS: Replication vs. Caching Strategies for Distributed Metadata Management in Cloud Storage Systems”, Proceedings of the Sixth ACM Symposium on Cloud Computing, Kohala Coast, HI, 2015, pp. 236-249.

[13] M. Abdul, M. Sharfuddin and T. Ragunathan, “Support-based replication algorithm for cloud storage systems”, Proceedings of the 7th ACM India Computing Conference, Nagpur, IN, 2014, pp. 1-9.

[14] K. A. Nuaimi, N. Mohamed, M. A. Nuaimi and J. A. Jaroodi, "A Novel Approach for Dual-Direction Load Balancing and Storage Optimization in Cloud Services," 2014 IEEE 13th International Symposium on Network Computing and Applications, Cambridge, MA, 2014, pp. 105-112.

[15] A. Stiemer, I. Fetai and H. Schuldt, "Analyzing the performance of data replication and data partitioning in the cloud: The BEOWULF approach," 2016 IEEE International Conference on Big Data (Big Data), Washington, DC, 2016, pp. 2837-2846.

[16] N. Tziritas et al., "Data Replication and Virtual Machine Migrations to Mitigate Network Overhead in Edge Computing Systems," in IEEE Transactions on Sustainable Computing, vol. 2, no. 4, pp. 320-332, 1 Oct.-Dec. 2017.

[17] S. D. Salunkhe and D. Patil, "Division and replication for data with public auditing scheme for cloud storage," 2016 International Conference on Computing Communication Control and automation (ICCUBEA), Pune, 2016, pp. 1-5.

[18] M. Ali, K. Bilal, S. U. Khan, B. Veeravalli, K. Li and A. Y. Zomaya, "DROPS: Division and Replication of Data in Cloud for Optimal Performance and Security," in IEEE Transactions on Cloud Computing, vol. 6, no. 2, pp. 303-315, 1 April-June 2018.

[19] K. Al Nuaimi, N. Mohamed, M. Al Nuaimi and J. Al-Jaroodi, "Dual direction load balancing and partial replication storage of cloud DaaS," 2014 IEEE 3rd International Conference on Cloud Networking (CloudNet), Luxembourg, 2014, pp. 432-437.

[20] Ching-Yang Tseng, ChangChun Lu and Cheng-Fu Chou, "Efficient privacy-preserving multi-keyword ranked search utilizing document replication and partition," 2015 12th Annual IEEE Consumer

Communications and Networking Conference (CCNC), Las Vegas, NV, 2015, pp. 671-676.

[21] B. Zhu, H. Li, K. W. Shum and S. R. Li, "HFR code: a flexible replication scheme for cloud storage systems," in IET Communications, vol. 9, no. 17, pp. 2095-2100, 26 11 2015.

[22] A. Mseddi, M. A. Salahuddin, M. F. Zhani, H. Elbiaze and R. H. Glitho, "On optimizing replica migration in distributed cloud storage systems," 2015 IEEE 4th International Conference on Cloud Networking (CloudNet), Niagara Falls, ON, 2015, pp. 191-197.

[23] R. J. Thara, S. Shine and C. Sanu, "Optimizing the performance of Database as a Service (DaaS) model — A distributed approach," 2013 Fourth International Conference on Computing, Communications and Networking Technologies (ICCCNT), Tiruchengode, 2013, pp. 1-5.

[24] Dezhi Hou, Shidong Zhang and Lanju Kong, "Placement of SaaS cloud data and dynamically access scheduling strategy," 2013 8th International Conference on Computer Science & Education, Colombo, 2013, pp. 834-838.

[25] J. Li and M. Hou, "PMSTCOM: A Novel Replication Consistency Maintenance Strategy in Cloud Storage System," 2017 3rd International Conference on Big Data Computing and Communications (BIGCOM), Chengdu, 2017, pp. 277-283.

[26] A. Tripathi and G. Rajappan, "Scalable Transaction Management for Partially Replicated Data in Cloud Computing Environments," 2016 IEEE 9th International Conference on Cloud Computing (CLOUD), San Francisco, CA, 2016, pp. 260-267.

[27] Yun Tian, Xiao Qin and Yafei Jia, "Secure replica allocation in cloud storage systems with heterogeneous vulnerabilities," 2015 IEEE International Conference on Networking, Architecture and Storage (NAS), Boston, MA, 2015, pp. 205-214.

[28] X. Luo, Y. Wang, Z. Zhang and H. Wang, "Superset: A Non-uniform Replica Placement Strategy towards High-Performance and Cost-Effective Distributed Storage Service," 2013 International Conference on Advanced Cloud and Big Data, Nanjing, 2013, pp. 139-146.

[29] F. Xie, J. Yan and J. Shen, "Towards Cost Reduction in Cloud-Based Workflow Management through Data Replication," 2017 Fifth International Conference on Advanced Cloud and Big Data (CBD), Shanghai, 2017, pp. 94-99.

[30] C. Hsu, V. W. Freeh and F. Villanustre, "Trilogy: Data placement to improve performance and robustness of cloud computing," 2017 IEEE International Conference on Big Data (Big Data), Boston, MA, 2017, pp. 2442-2451.

[31] B. Alami, N. Jafari, "A comprehensive review of the data replication techniques in the cloud environments: Major trends and future directions", Journal of Network and Computer Applications, Vol. 64, pp. 229-238, 2016.

[32] T. Hamrouni, S. Slimani and F. Ben, "A data mining correlated patterns-based periodic decentralized replication strategy for data grids", Journal of Systems and Software, Vol. 110, pp. 10-27, 2015.

[33] N. Kaur and S. Singh, "A dynamic, cost-aware, optimized data replication strategy for heterogeneous cloud data centers", Future Generation Computer Systems, Vol. 65, pp. 10-32, 2016.

[34] N. Mansouri and M. M. Javidi, "A new Prefetching-aware Data Replication to decrease access latency in cloud environment", Journal of Systems and Software, Vol. 149, pp. 197-215, 2018.

[35] A. A. Amer, A. A. Sewisy, T. M. A. Elgendy, "An optimized approach for simultaneous horizontal data fragmentation and allocation in Distributed Database Systems (DDBSs)", Heliyon, Vol. 3, pp. 1-42, 2017.

[36] A. Brigit, "Data allocation optimization for query processing in graph databases using Lucene", Computers & Electrical Engineering, Vol. 70, pp. 1019-1033, 2018.

[37] G. Montoya, H. Skaf-Molli, P. Molli and M. Vidal,"Decomposing federated queries in presence of replicated fragments", Web Semantics: Science, Services and Agents on the World Wide Web, Vol. 42, pp. 1-18, 2017.

[38] N. Mansouri and M. Kuchaki, M.M. Javidi, "DPRS: A dynamic popularity aware replication strategy with parallel download scheme in cloud environments", Simulation Modelling Practice and Theory, Vol. 77, pp. 177-196, 2017.

[39] C. Rosas, A. Sikora, J. Jorba, A. Moreno, A. Espinosa and E. César, "Dynamic tuning of the workload partition factor and the resource utilization in data-intensive applications", Future Generation Computer Systems, Vol. 37, pp. 162-177, 2014.

[40] H. Fan, Z. Ma, D. Wang and J. Liu,"Handling distributed XML queries over large XML data based on MapReduce framework", Information Sciences, Vol. 453, pp. 1-20, 2018.

[41] L. Gkatzikis, V. Sourlas, C. Fischione and I. Koutsopoulos, "Low complexity content replication through clustering in Content-Delivery Networks", Computer Networks, Vol. 121, pp. 137-151, 2017.

[42] S. Long, Y. Zhao and W. Chen, "MORM: A Multi-objective Optimized Replication Management strategy for cloud storage cluster", Journal of Systems Architecture, Vol. 60, pp. 234-244, 2014.

[43] S. Sengupta and K.M. Annervaz, "Multi-site data distribution for disaster recovery—A planning framework", Future Generation Computer Systems, Vol. 41, pp. 53-64, 2014.

[44] Z. Zeng and B. Veeravalli, "Optimal metadata replications and request balancing strategy on cloud data centers", Journal of Parallel and Distributed Computing, Vol. 74, pp. 2934-2940, 2014.

[45] T.P. Shabeera, S.D. Madhu, S. M. Salam and K. Murali, "Optimizing VM allocation and data placement for data-intensive applications in cloud using ACO metaheuristic algorithm", Engineering Science and Technology, an International Journal, Vol. 20, pp. 616-628, 2017.

[46] T. Habib and Z. Ansari, "Partition based clustering of large datasets using MapReduce framework: An analysis of recent themes and directions", Future Computing and Informatics Journal, Vol. 20, pp. 1-15, 2018.

[47] D. Sánchez and M. Batet, "Privacy-preserving data outsourcing in the cloud via semantic data splitting", Computer Communications, Vol. 110, pp. 187-201, 2017.

[48] T. Chen, R. Bahsoon and A. H. Tawil, "Scalable service-oriented replication with flexible consistency guarantee in the cloud", Information Sciences, Vol. 264, pp. 349-370, 2014.

[49] Q. Xia, W. Liang and Z. Xu, "The operational cost minimization in distributed clouds via community-aware user data placements of social networks", Computer Networks, Vol. 112, pp. 263-278, 2017.

[50] O. Romero, V. Herrero, A. Abelló and J. Ferrarons, "Tuning small analytics on Big Data: Data partitioning and secondary indexes in the Hadoop ecosystem", Information Systems, Vol. 54, pp. 336-356, 2015.

[51] S. Benkrid, L. Bellatreche, A. Cuzzocrea, "A Global Paradigm for Designing Parallel Relational Data Warehouses in Distributed Environments", Transactions on Large-Scale Data- and Knowledge-Centered Systems XV, Vol. 8920, pp. 64-101, 2014.

[52] N. Mansouri and M. M. Javidi, "A hybrid data replication strategy with fuzzy-based deletion for heterogeneous cloud data centers", The Journal of Supercomputing (2018), Vol. 74, pṕ. 1-24, 2018.

[53] A. Guabtni, R. Ranjan, F. A. Rabhi, "A workload-driven approach to database query processing in the cloud", The Journal of Supercomputing, Vol. 63, pp. 722–736, 2013.

[54] A. S. Darabant, L. Tambulea, V. Varga, "Access Patterns Optimization in Distributed Databases Using Data Reallocation", Database and Expert Systems Applications, Vol. 10438, pp. 178-186, 2017.

[55] N. Mansouri, "Adaptive data replication strategy in cloud computing for performance improvement", Frontiers of Computer Science, Vol. 10, pp. 925–935, 2016.

[56] E. B. Edwin, P. Umamaheswari, M. R. Thanka, "An efficient and improved multi-objective optimized replication management with dynamic and cost aware strategies in cloud computing data center", Cluster Computing, Vol. 21, pp. 1-10, 2017.

[57] A. E. Abdel, N. L. Badr, M. F. Tolba, "An Enhanced Distributed Database Design Over the Cloud Environment", Proceedings of the International Conference on Advanced Intelligent Systems and Informatics 2016, Vol. 533, pp. 290-298, 2016.

[58] L. Wiese, "Clustering-based fragmentation and data replication for flexible query answering in distributed databases", Journal of Cloud Computing, Vol. 3, pp. 1-15, 2014.

[59] K. Grolinger, W. A. Higashino , A. Tiwari and M. Capretz, "Data management in cloud environments: NoSQL and NewSQL data stores", Journal of Cloud Computing, Vol. 2, pp. 1-24, 2013.

[60] J. Mohammad and M. Murshed, “Distributed Database Management Systems: Architectural Design Choices for the Cloud”, Cloud Computing, Vol. 7, pp. 23-50, 2014.

[61] A. E. Abdel, N. L. Badr and M. F. Tolba, “Distributed Database System (DSS) Design Over a Cloud Environment”, Multimedia Forensics and Security, Vol. 115, pp. 97-116, 2016.

[62] A. E. Abdel, N. L. Badr, and M. F. Tolba, “Dynamic Distributed Database over Cloud Environment”, AMLTA 2014: Advanced Machine Learning Technologies and Applications, Vol. 488, pp. 67-76, 2014.

[63] A. V. Kish, J. R. Rose, and C. Farkas, “Efficient Partitioning and Allocation of Data for Workload Queries”, Innovations and Advances in Computing, Informatics, Systems Sciences, Networking and Engineering, Vol. 313, pp. 549-555, 2014.

[64] X. Zou, J. Pan, W. Du, and S. Chen, “Elastic Database Replication in the Cloud”, ICA3PP 2015: Algorithms and Architectures for Parallel Processing, Vol. 9531, pp. 667-681, 2015.

[65] E. Insaf and G. Belalem, “Optimization of Tasks Scheduling by an Efficacy Data Placement and Replication in Cloud Computing”, ICA3PP 2013: Algorithms and Architectures for Parallel Processing, Vol. 8286, pp. 22-29, 2013.

[66] Z. Gao, H. Min, X. Li, J. Huang, Y. Jin, A. Lei, S. Bourbonnais, M. Zheng, G. Fuh, “Optimizing Inter-data-center Large-Scale Database Parallel Replication with Workload-Driven Partitioning”, Transactions on Large-Scale Data- and Knowledge-Centered Systems XXIV, Vol. 9510, pp. 169-192, 2016.

[67] S. U. Rehman, S. U. Khan, S. J. Ewen, N. Tziritas, J. Kolodziej, A. Y. Zomaya, S. A. Madani, N. Min-Allah, L. Wang, C. Xu, Q. M. Malluhi, J. E. Pecero, P. Balaji, A. Vishnu, R. Ranjan, S. Zeadally, H. Li, “Performance analysis of data intensive cloud systems based on data management and replication: a survey”, Distributed and Parallel Databases, Vol. 34, pp. 179–215, 2016.

[68] L. Bellatreche, A.Kerkad, S. Breß, D. Geniet, “RouPar: Routinely and Mixed Query-Driven Approach for Data Partitioning” , OTM Confederated International Conferences "On the Move to Meaningful Internet Systems", Vol. 8185, pp. 309-326, 2013.

[69] S. Ahirrao, R. Ingle, “Scalable transactions in cloud data stores”, Journal of Cloud Computing, Vol. 4, pp. 1-14, 2015.

[70] F. Hassen, A. G. Touzi, “Towards a New Architecture for the Description and Manipulation of Large Distributed Data”, Big Data in Complex Systems, Vol. 9, pp. 471-497, 2015.

[71] W. Lehner, K. Sattler, “Transactional Data Management Services for the Cloud”, Web-Scale Data Management for the Cloud, Vol. 1, pp. 59-90, 2013.

[72] N. Phaphoom, X. Wang and P Abrahamsson, “Foundations and Technological Landscape of Cloud Computing”, ISRN Software Engineering, vol. 2013, 2013, pp. 1-31.

[73] L. Wiese, T. Waage and F. Bollwein, "A Replication Scheme for Multiple Fragmentations with Overlapping Fragments," in The Computer Journal, vol. 60, no. 3, pp. 308-328, March 2017.

[74] A. Noraziah, A. Azila, A. Fairuzullah, N. Aqtar, A. Zinuddin and T. Herawan, “Fault Tolerance On Binary Vote Assignment Cloud Quorum (BVACQ) Replication Technique”, Proceeding of the 2013 International Conference on Computer Science and Information Technology (CSIT 2013), Jogjakarta, ID, 2013, pp. 1-6.

[75] A. A. Che Fauzi et al., "Managing Fragmented Database Replication for Mygrants Using Binary Vote Assignment on Cloud Quorum", Applied Mechanics and Materials, Vols. 490-491, pp. 1342-1346, 2014.

[76] A. A. El Mrabti, N. Ammari, A. A. El Kalam, A. A. Ouahman and M. De Montfort, “New mechanism for Cloud Computing Storage Security” International Journal of Advanced Computer Science and Applications(IJACSA), Vol. 7, pp. 526-539, 2016.

[77] V. Khandelwal, “Secure and Efficient Data Storage in Multi-clouds”, International Journal of Information and Computation Technology, Vol. 3 no. 9, pp. 977-984, 2013.

[78] S. Singh and R. Sapra, “Secure Replication management in cloud storage”, International Journal of Emerging Trends & Technology in Computer Science (IJETTCS), Vol. 3 no. 2, pp. 251-254, 2014.

[79] S. S and K. V, "A Metadata Based Storage Model for Securing Data in Cloud Environment," 2011 International Conference on Cyber-Enabled Distributed Computing and Knowledge Discovery, Beijing, 2011, pp. 429-434.

[80] M. Scheidgen, A. Zubow, J. Fischer, T. H. Kolbe, “Automated and Transparent Model Fragmentation for Persisting Large Models”, International Conference on Model Driven Engineering Languages and Systems, Vol. 7590, pp. 102-118, 2012.

[81] J. O. Hauglid, N. H. Ryeng, K. Nørvåg, “DYFRAM: dynamic fragmentation and replica management in distributed database systems”, Distributed and Parallel Databases, Vol. 28, pp. 157–185, 2010.

[82] S. Das, D. Agrawal, A. E. Abbadi, “ElasTraS: An Elastic, Scalable, and Self Managing Transactional Database for the Cloud”, ACM Transactions on Database Systems, Vol. 38, pp. 1-45, 2013.

[83] L. Wiese, “Horizontal Fragmentation and Replication for Multiple Relaxation Attributes”, British International Conference on Databases, Vol. 9147, pp. 157-169, 2015.

[84] D. Bermbach, M. Klems, S. Tai and M. Menzel, "MetaStorage: A Federated Cloud Storage System to Manage Consistency-Latency Tradeoffs," 2011 IEEE 4th International Conference on Cloud Computing, Washington, DC, 2011, pp. 452-459.

[85] A. F. Barsoum and M. A. Hasan, “On Verifying Dynamic Multiple Data Copies over Cloud Servers”, International Association for Cryptologic Research Cryptology ePrint Archive, Vol. 2011, pp. 1-30, 2011.

[86] Oracle Corporation (2018, Oct 13). JavaServer Faces Technology [Online]. Available: https://www.oracle.com/technetwork/java/javaee/javaserverfaces-139869.html

[87] Amazon Web Services (2018, Oct 9). Cloud Computing with Amazon Web Services [Online]. Available: https://aws.amazon.com/es/what-is-aws/

[88] Oracle Corporation (2018, Oct 13). NetBeans IDE [Online]. Available: https://netbeans.org/

[89] Institute for Informatics Research Unit of Programming and Software Engineering (2018, Oct 13). UWE – UML-based Web Engineering [Online]. Available: http://uwe.pst.ifi.lmu.de/

[90] Oracle Corporation (2018, Oct 13). MySQL [Online]. Available: https://www.mysql.com

[91] K. Bahwaireth, L. Tawalbeh, E. Benkhelifa, Y. Jararweh and M. A. Tawalbeh, “Experimental comparison of simulation tools for efficient cloud and mobile cloud computing applications”, EURASIP Journal on Information Security, 2016, vol. 2016, pp. 1-15.

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